

FinWise AI - Hackathon Submission Document

Problem Statement

College students and young adults frequently face a severe lack of practical financial literacy, leading to compounding debt, poor budgeting habits, and an inability to understand the long-term impact of educational loans. Traditional banking applications are often overly complex, intimidating, and lack the educational context required to help students make informed, proactive financial decisions. Furthermore, standard financial tools fail to provide hyper-personalized, contextual advice based on a user's unique income and debt realities.

Solution Statement

FinWise AI solves this by serving as an intelligent, gamified financial copilot tailored specifically for students. It abstracts complex financial mathematics-such as Amortization schedules, compound interest, and affordability ratios-into an intuitive, highly visual dashboard. By enforcing a strict 'Financial Safety Principle,' our FastAPI backend performs all authoritative calculations, ensuring mathematical accuracy and security.

We combine proactive tools (like dynamic budgets and savings goals) with a powerful AI Copilot. Instead of generic advice, our platform serializes the user's exact database metrics-including their proprietary Financial Health Score and active loan EMIs-and feeds them into the AI engine. This allows FinWise AI to deliver mathematically precise, hyper-personalized financial guidance, transforming overwhelming banking data into actionable, educational insights.

Technology Used - IBM Bob

IBM Bob was utilized as the central cognitive engine and development accelerator for the FinWise AI platform, fundamentally shaping both the architecture and the core user experience.

1. The AI Copilot (Core Feature Integration):

The most prominent implementation of IBM Bob is within the platform's 'AI Copilot' feature. Standard financial apps provide numbers without context; IBM Bob acts as the contextual bridge. When a student asks a question like 'Why is my financial health score failing?', the FastAPI backend securely serializes the user's actual PostgreSQL data-including their proprietary 6-axis Financial Health Score, wallet balance, and active budget deficits. This raw data is injected into the System Prompt of the IBM Bob LLM endpoint. Because IBM Bob processes this highly contextual prompt, it generates personalized, mathematically accurate advice (e.g., 'You must reduce your Food budget by \$30 to comfortably afford your upcoming EMI'). IBM Bob transforms static database rows into an interactive, educational dialogue.

2. Algorithm Generation & Validation:

Behind the scenes, IBM Bob technology was heavily utilized during the development phase to architect and validate our complex financial algorithms. Developing accurate, edge-case resilient formulas for reducing-balance EMIs, compound interest calculations, and the weighted 100-point Financial Health Score

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required immense precision. IBM Bob assisted in generating the core Python mathematical models, specifically accounting for standard financial rules like 'Banker's Rounding' (IEEE 754), ensuring that the backend logic is enterprise-grade and mathematically sound.

3. Workflow Optimization:

We integrated IBM Bob into our development workflow to streamline the decoupling of our architecture. By acting as a technical co-pilot, IBM Bob guided the strict separation of concerns-ensuring the React frontend strictly handled state management and UI caching, while the FastAPI backend handled all secure database transactions (ACID compliance) and auth token generation. This allowed us to build a robust, scalable product at hackathon speed without compromising on security or code quality.